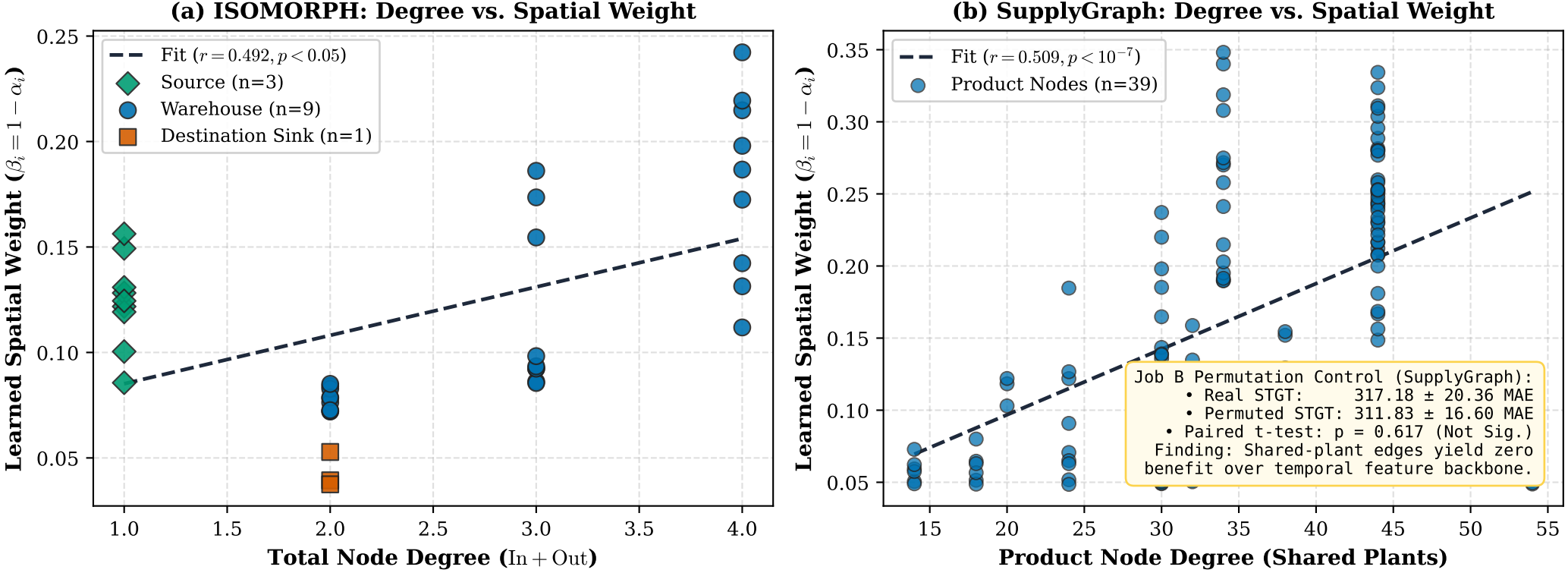

(a) ISOMORPH: Degree vs. Spatial Weight
Fit ($r = 0.492, p < 0.05$)
Source (n=3)
Warehouse (n=9)
Destination Sink (n=1)
Learned Spatial Weight ($\beta_i = 1 - \alpha_i$)
Total Node Degree (In + Out)
0.25
0.20
0.15
0.10
0.05
1.0
1.5
2.0
2.5
3.0
3.5
4.0
(b) SupplyGraph: Degree vs. Spatial Weight
Fit ($r = 0.509, p < 10^{-7}$)
Product Nodes (n=39)
Learned Spatial Weight ($\beta_i = 1 - \alpha_i$)
Product Node Degree (Shared Plants)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
15
20
25
30
35
40
45
50
55
Job B Permutation Control (SupplyGraph):
• Real STGT: 317.18 ± 20.36 MAE
• Permuted STGT: 311.83 ± 16.60 MAE
• Paired t-test: p = 0.617 (Not Sig.)
Finding: Shared-plant edges yield zero benefit over temporal feature backbone.